ELEC3104: Digital Signal Processing

Topic 4: Digital Resonator Fundamentals

- Second-Order Digital Resonator (pp. 3-8)
- Stability of a Second - Order Filter (pp. 9-11)
- Digital Oscillators (pp. 12-19)
- Tutorial questions (pp. 20-25)

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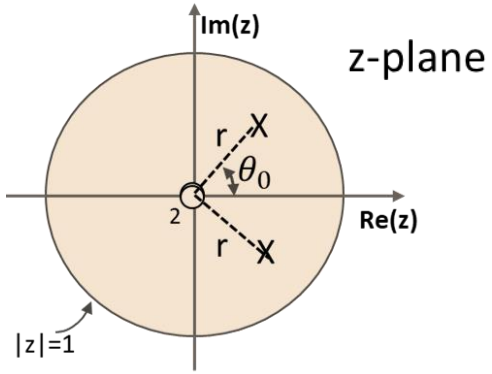
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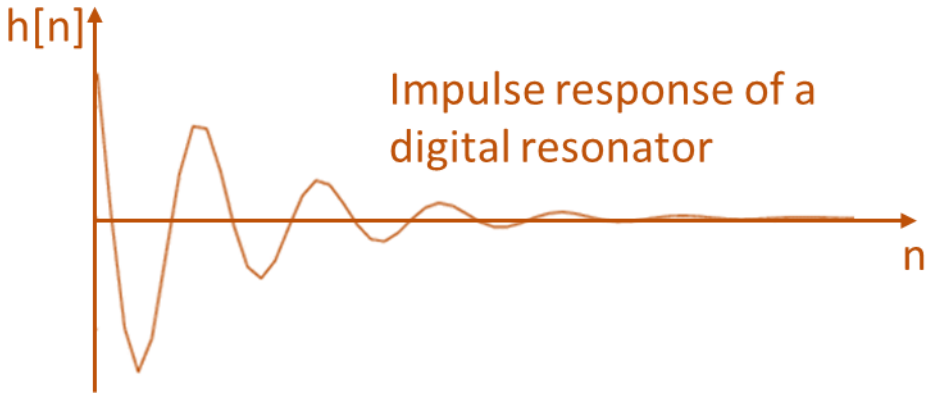
Digital Resonator Fundamentals

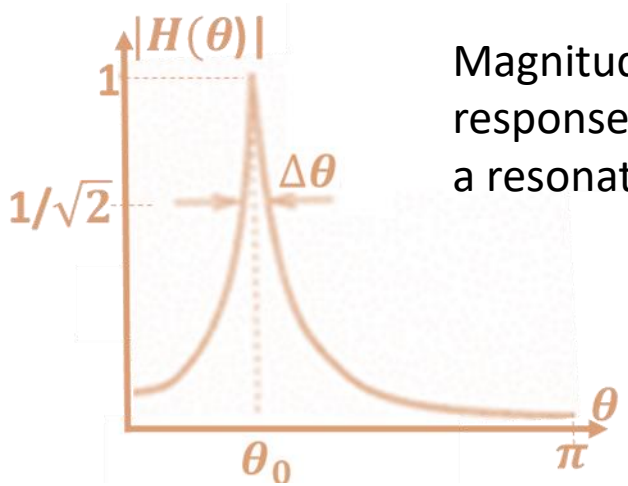
Introduction

- A digital resonator is normally a two-pole IIR filter having a complex conjugate pair of poles located inside the unit circle of the z-plane and a pair of zeros at z=0.
- The angle of the poles in polar co-ordinates determines the resonant frequency θ_0 of the resonator, while the distance (r) of the poles from the origin of the unit circle sets the bandwidth $\Delta\theta$.
- The conjugate poles closer to the unit circle ensure smaller bandwidth, but with a sharper peak at the resonant frequency θ_0 .
- The digital resonator is essentially a narrow-bandpass filter.









Magnitude response of a resonator

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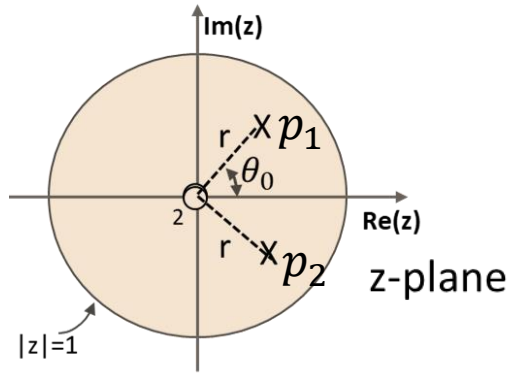
Digital Resonator Fundamentals: A second-order digital resonator

A second-order digital resonator

- One way to realise a second-order resonator with a peak at θ_0 is to place a pair of poles with the frequency of $\pm\theta_0$ ($p_1 = re^{j\theta_0}$ and $p_2 = re^{-j\theta_0}$) and a pair of zeros at z=0. (see diagram on the right)
- To ensure stability , we choose the pole radius r to be less than (but close to) unity.
- The transfer function of such a digital resonator with zeros at the origin is:

$$H(z) = \frac{z^2}{(z-p_1)(z-p_2)} = \frac{z^2}{(z-re^{j\theta_0})(z-re^{-j\theta_0})}$$

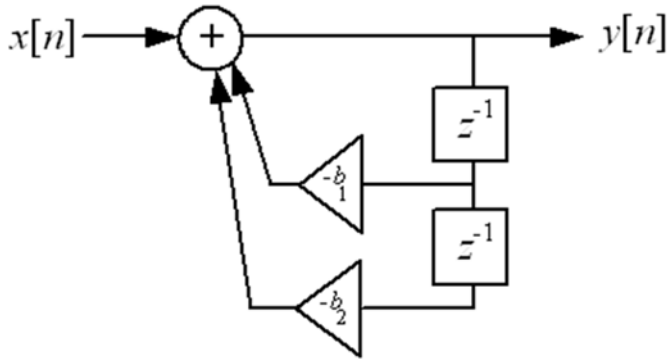
$$H(z) = \frac{z^2}{z^2 - r(e^{j\theta_0}+e^{-j\theta_0})z + r^2} = \frac{z^2}{z^2 - 2r\cos\theta_0 z + r^2} \quad \text{..... (1)}$$



$$p_1 = re^{j\theta_0} = r\cos\theta_0 + jr\sin\theta_0$$
$$p_2 = re^{-j\theta_0} = r\cos\theta_0 - jr\sin\theta_0$$

From the second order realisation (see diagram on the right), we have

$$H(z) = \frac{1}{1 + b_1 z^{-1} + b_2 z^{-2}} = \frac{z^2}{z^2 + b_1 z + b_2} \quad \text{.... (2)}$$



A second-order structure

Equations (1) and (2) are equivalent. By equating the coefficients, we obtain

$$b_1 = -2r\cos\theta_0 \quad \text{and} \quad b_2 = r^2; \quad \therefore \cos\theta_0 = -\frac{b_1}{2\sqrt{b_2}} \qquad \theta_0 = \frac{2\pi f_0}{f_s}$$

Digital Resonator Fundamentals

Magnitude Spectrum

- The digital resonator's magnitude squared function $|H(\theta)|^2$ can be obtained from H(z) as follows: $H(z) = \frac{z^2}{z^2 - 2r\cos\theta_0 z + r^2}$

$$|H(\theta)|^2 \left|\frac{e^{j2\theta}}{e^{j2\theta} - 2r\cos\theta_0 e^{j\theta} + r^2}\right|^2 = \frac{1}{[1-2r\cos(\theta-\theta_0)+r^2]\ [1-2r\cos(\theta+\theta_0)+r^2]}$$

- From the magnitude spectrum shown on the right, it can be seen, the peak value *A* occurs at $\theta = \theta_0$ (or very close to) the resonant frequency (θ_0) and is given by

$$A^2 = |H(\theta)|^2 = \frac{1}{[(1-r)^2]\ [1-2r\cos(2\theta_0)+r^2]}$$

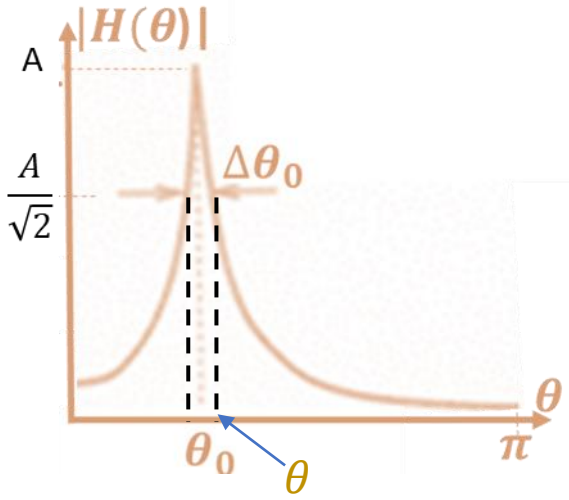
- The half-power bandwidth $\Delta\theta$ is found by locating the frequencies at which $|H(\theta)|^2 = 0.5\ |H(\theta_0)|^2 = 0.5\ A^2$.

$$\therefore \frac{1}{[1-2r\cos(\theta-\theta_0)+r^2]\ [1-2r\cos(\theta+\theta_0)+r^2]} = \frac{0.5}{[(1-r)^2]\ [1-2r\cos(2\theta_0)+r^2]}$$

- Since the half-power points are very close to θ_0, we have $\theta + \theta_0 \approx 2\theta_0$ and $\theta - \theta_0 \approx \frac{1}{2}\Delta\theta$ and the above relationship simplifies to

$$1 - 2r\cos(\tfrac{1}{2}\Delta\theta) + r^2 = 2(1-r)^2$$

- This relationship between the pole radius r and the bandwidth $\Delta\theta$ can be simplified when the poles are very close to the unit circle (with r > 0.9 or so). The values of *r* and the peak magnitude, A, are then reasonably approximated by:



The peak gain can be normalised to unity by dividing the transfer function *H(z)* by A.

$$r \approx 1 - 0.5\,\Delta\theta$$

$$\Delta\theta = 2\pi\frac{\Delta f}{f_s};\ r \approx 1 - \pi\frac{\Delta f}{f_s}$$

$$A \approx \frac{1}{(1-r^2)\sin(\theta_0)}$$

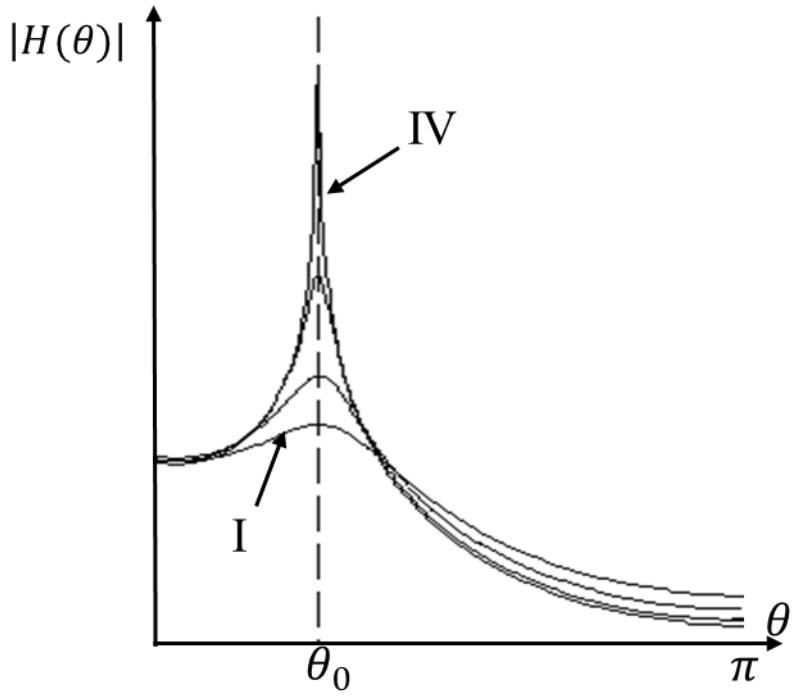
Digital Resonator Fundamentals

Magnitude Spectrum

- From the resonator equation H(z), we can derive the magnitude spectrum as discussed before:

$$H(z) = \frac{1}{1+b_1 z^{-1}+b_2 z^{-2}}; \quad b_2 = r^2; \quad b_1 = -2r\,cos(\theta_0); \quad \theta_0 = cos^{-1}\left[\frac{-b_1}{2\sqrt{b_2}}\right]$$

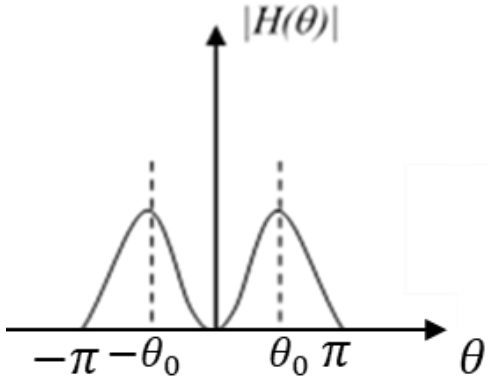
- The magnitude spectrum is shown below for various values of b_1 and b_2.



	b_1	b_2
I	-0.94	0.5
II	-1.16	0.7
III	-1.34	0.9
IV	-1.41	0.99

To null the magnitude response at low and high frequencies (θ=0 and θ= π) , the two zeros in H(z) may be located at z = 1 and z = -1 and this leads to the modified transfer function $H_1(z)$ and its magnitude response is shown below:

$$H_1(z) = \frac{1 - z^{-2}}{1 + b_1 z^{-1} + b_2 z^{-2}}$$



Magnitude response

Digital Resonator Fundamentals

Example 1 [Reference 2]:

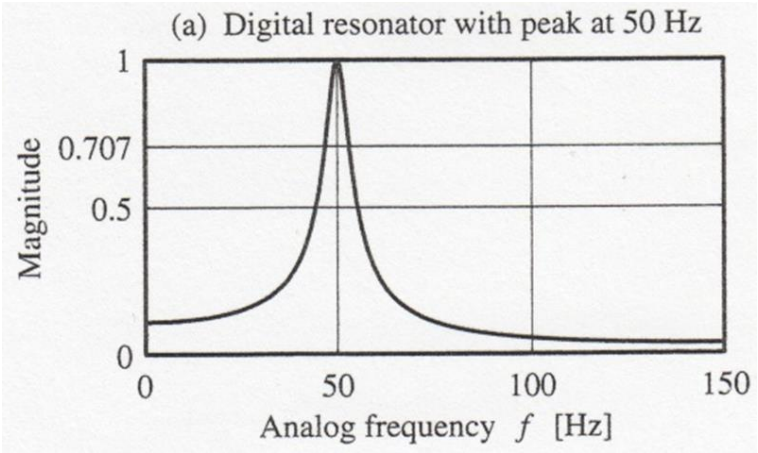
- Design a digital resonator with a peak gain of unity ay 50Hz and a 3-dB bandwidth of 6 Hz , assuming the sampling frequency of 300Hz.
- The resonant frequency $\theta_0 = \frac{2\pi(50)}{300} = \frac{\pi}{3}$;
- The 3-dB bandwidth $\Delta\theta = \frac{2\pi(6)}{300} = 0.04\pi$.
- The pole radius (r) = 1- 0.5 $\Delta\theta$ = 1-0.02 π = 0.9372
- The transfer function of the digital resonator H(z) is:

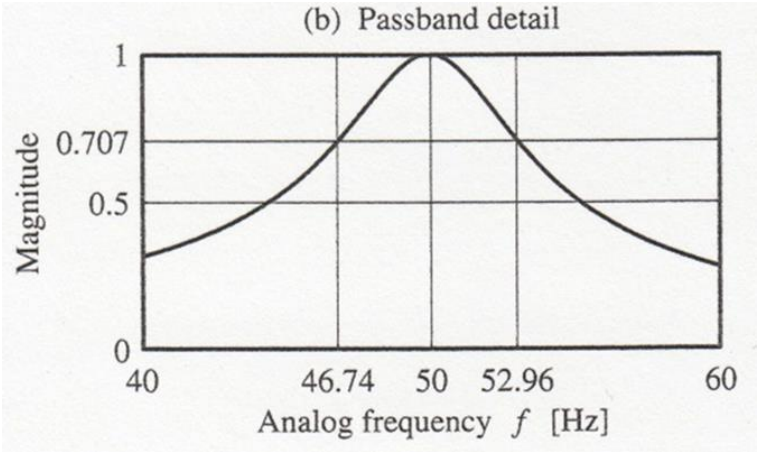
$$H(z) = \frac{k_0 z^2}{z^2 - 2r\cos\theta_0 z + r^2} = \frac{k_0 z^2}{z^2 - 0.9372z + 0.8783}$$

- For a peak gain of unity, we choose $k_0 = \frac{1}{A} = (1 - r^2)\sin(\theta_0) = 0.1054$

$$\therefore H(z) = \frac{0.1054 z^2}{z^2 - 0.9372z + 0.8783} = \frac{0.1054}{1 - 0.9372z^{-1} + 0.8783z^{-2}}$$

- The magnitude spectrum and the passband details are shown on the right
- The passband details shows that the 3 dB frequencies are located at 46.74 Hz and 52.96 Hz and the bandwidth is = (52.96) – (46.74) ≈ 6Hz





Digital Resonator Fundamentals : Notch filters

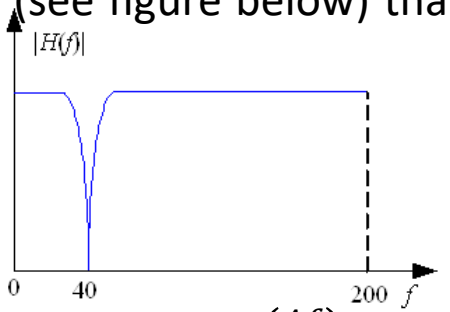
Notch filters

- When a zero is placed at a given point on the unit circle, the frequency response will be zero at the corresponding point.
- A pole on the other hand produces a peak at the corresponding frequency point.
- Poles that are close to the unit circle give rise large peaks, whereas zeros close to or on the unit circle produces troughs or minima.
- Thus, by strategically placing poles and zeros on the z-plane, we can obtain simple low pass or other frequency selective filters (**notch filters**).

Example 2:

Obtain, by the pole-zero placement method, the transfer function of a sample digital notch filter (see figure below) that meets the following specifications:

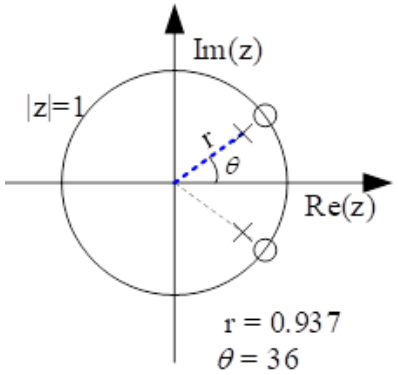
- Notch Frequency: 40Hz
- 3db width of the Notch: ± 4Hz
- Sampling frequency: 400 Hz



The radius, r of the poles is determined by: $r = 1 - \left(\frac{\Delta f}{f_s}\right)\pi$

- To reject the component at 40 Hz, place a pair of complex zeros at the points on the unit circle corresponding to ± 40 Hz. i.e. at an angles: $360^{\bullet} \times \frac{40}{400} = \pm 36^{\bullet} = \pm 0.2\pi$
- To achieve a sharp notch filter and improved magnitude response either side of the notch frequency, a pair of complex conjugate poles are placed at a radius $r < 1$, at the same frequency as the zeroes, as shown below.
- $r = 1 - \left(\frac{\Delta f}{f_s}\right)\pi = 1 - \left(\frac{8}{400}\right)\pi = 0.937$

$$H(z) = \frac{\left(z - e^{-j0.2\pi}\right)\left(z - e^{j0.2\pi}\right)}{\left(z - 0.937e^{-j0.2\pi}\right)\left(z - 0.937e^{j0.2\pi}\right)}$$



$$H(z) = \frac{1 - 1.6180z^{-1} + z^{-2}}{1 - 1.5161z^{-1} + 0.878z^{-2}}$$

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Digital Resonator Fundamentals : Stability of a Second-Order Filter

Stability

- Consider a two-pole resonant filter given by:

$$H(z) = \frac{z^2}{z^2 + b_1 z + b_2} = \frac{1}{1 + b_1 z^{-1} + b_2 z^{-2}}$$

where b_1 & b_2 are the coefficients.

- This system has two zeros at the origin and poles at

$$p_1, p_2 = -\frac{b_1}{2} \pm \frac{\sqrt{{b_1}^2 - 4b_2}}{2}$$

- The filter is stable if the poles lie inside the unit circle i.e.

$|p_1| < 1$ & $|p_2| < 1$

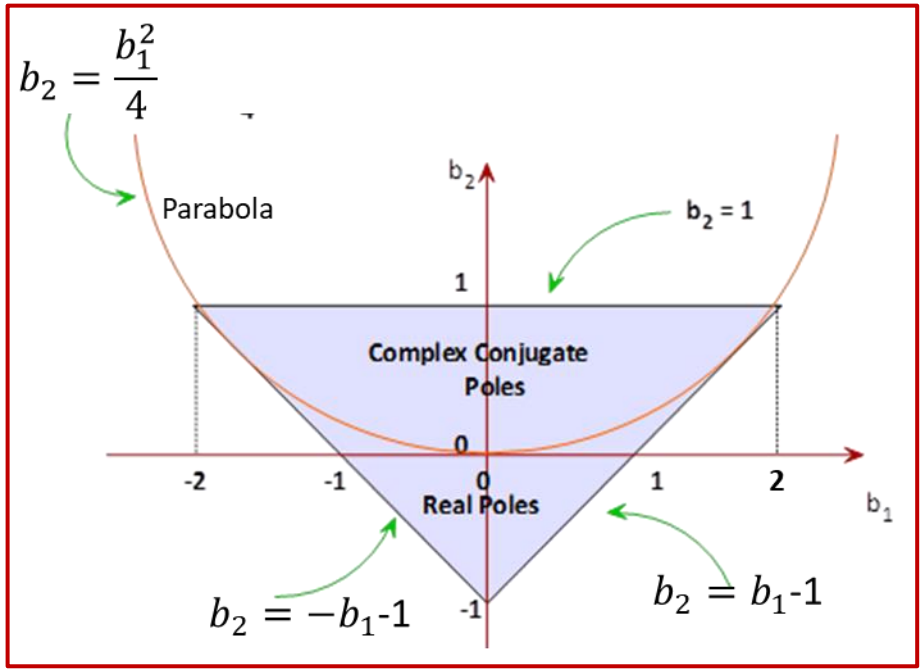
- For stability $b_2 < 1$. If $b_2 = 1$ then the system is an oscillator (**Marginally stable**)
- Assume that the poles are complex:

$${b_1}^2 - 4b_2 < 0 \quad \Rightarrow \quad {b_1}^2 < 4b_2 \quad \text{and} \quad b_1 < \pm 2\sqrt{b_2} \qquad b_2 > 0$$

$$b_1 = -2r\cos\theta_0 \quad \text{and} \quad b_2 = r^2$$

- If ${b_1}^2 - 4b_2 \geq 0$ then we get real roots.
- The stability conditions define a region in the coefficient plane (b_1, b_2) which is in the form of a triangle (see diagram on the right)

The system is only stable if and only if the point (b_1, b_2) lies inside the **stability triangle**.



Digital Resonator Fundamentals : Stability of a Second-Order Filter

Stability Triangle

- If the two poles (p_1 *and* p_2) are real, then they must have a value between -1 and 1 for the system to be stable.

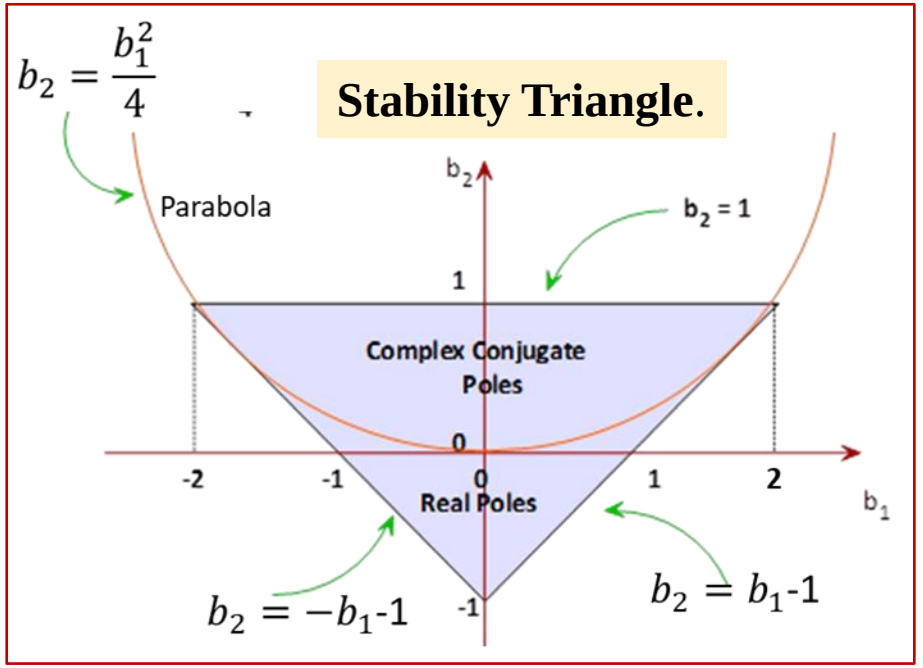
$$-1 < \frac{-b_1 \pm \sqrt{b_1^2 - 4b_2}}{2} < 1 \quad \Rightarrow \quad -2 + b_1 < \pm\sqrt{b_1^2 - 4b_2} < 2 + b_1$$

$$\therefore -2 + b_1 < -\sqrt{b_1^2 - 4b_2} \quad \text{and} \quad \sqrt{b_1^2 - 4b_2} < 2 + b_1$$

$$(-2 + b_1)^2 > b_1^2 - 4b_2 \quad \text{and} \quad b_1^2 - 4b_2 < (2 + b_1)^2$$

$$b_1 - b_2 - 1 < 0 \quad \text{and} \quad b_1 + b_2 + 1 > 0$$

- The region below the parabola ($b_1^2 > 4b_2$) corresponds to real and distinct poles.
- The points on the parabola ($b_1^2 = 4b_2$) result in real and equal (double) poles.
- The points above the parabola correspond to complex-conjugate poles.



For a stable 2nd order system, the coefficients must satisfy the following inequality:

$$-2 \leq b_1 \leq 2$$
$$-1 \leq b_1 \leq 1$$

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Digital Resonator Fundamentals : Digital Oscillators

Digital Oscillators

- A digital sinusoidal oscillator may be viewed as a limiting form of a two-pole resonator for which the complex-conjugate poles lie on the unit circle (see the pole-zero plot on the right)
- The impulse response, h[n], of the second-order filter with complex conjugate **poles on the unit circle** is a sinusoid and the system is called a digital sinusoid oscillator.

$$h[n] = A\ r^n \cos[n\theta] = A\ cos[n\theta]$$ [poles on the unit circle, r=1]

- From the z-transforms table, the transfer function of a causal oscillator that generates a pure sine wave $h[n] = A\ cos[n\theta]$ is given by ,

$$H(z) = \frac{Y(z)}{X(z)} = \frac{1-(cos\ \theta)\ z^{-1}}{1-(2\ cos\ \theta)\ z^{-1}+z^{-2}}$$

- Taking inverse z-transform on both sides, we obtain

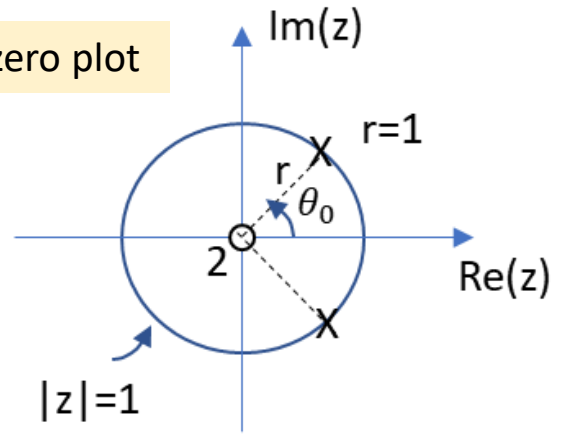
$$y[n] - 2\ cos\ \theta\ \ y[n-1] + y[n-2] = x[n] - cos\ \theta\ \ x[n-1]$$

- For an oscillator, the input x[n]=0 and x[n-1] =0; So, the equation of the digital oscillator becomes:

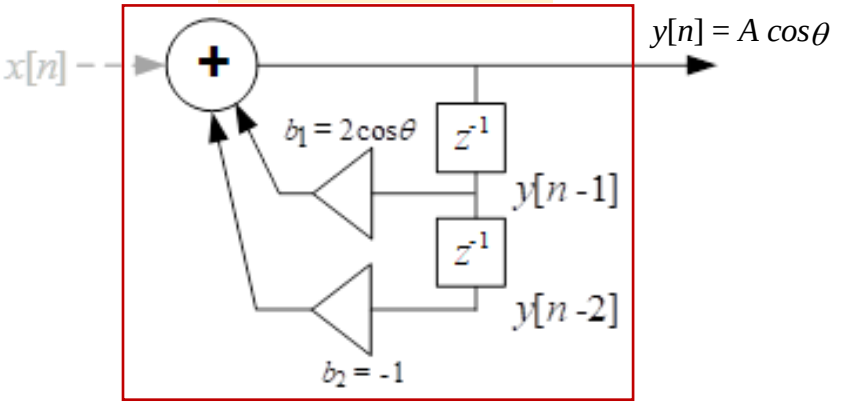
$$y[n] = 2\ cos\ \theta\ y[n-1] - y[n-2]$$

and its structure is shown below:

Pole-zero plot



Digital Oscillator



Digital Resonator Fundamentals : Digital Oscillators

Initial conditions

Case 1: Assume the input x[n] = 0:

- To obtain $y[n] = A\cos(n\theta)$, use the following initial conditions

$$y[0] = A\cos(0.\theta) = A \quad [for\ n=0]; \quad y[-1] = A\cos(-1.\theta) = A\cos\theta \quad [for\ n= -1]$$

- The frequency of the oscillator can be tuned by changing the coefficient b_1 (b_2 is a constant). The frequency θ of the oscillator is,

$$cos\,\theta = \frac{-b_1}{2\sqrt{b_2}} = -\frac{b_1}{2} \quad \text{[For an oscillator } b_2\text{=1]}$$

Case 2: Assume y[0]= 0 and y[-1]=0:

- In order to begin the sinusoidal oscillation, apply an impulse $\delta[n]$ at the input (see Figure A) when n= 0; Thereafter, the oscillation is self-sustaining because the system has no damping (i.e. r =1)

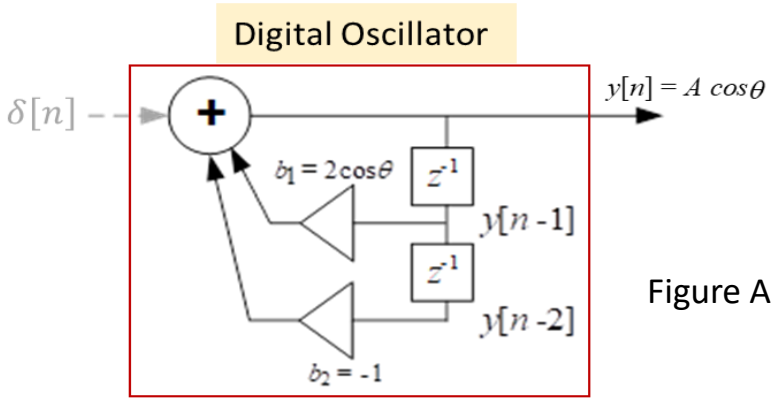


Figure A

Digital Oscillator Applications

- The touch tone phone (DTMF phone: Dual tone multi frequency) makes use of digital oscillators to generate audible tones by pressing buttons on a keypad as shown in Figure B.
- Pressing a button produces a two-tone signal containing, a high and a low frequency tone
- Each button is associated with a unique pair of low-frequency and high-frequency tones. The tones can be generated by using a parallel combination of two programmable digital oscillators

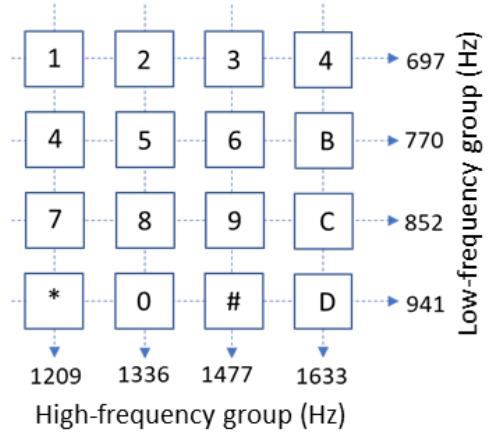
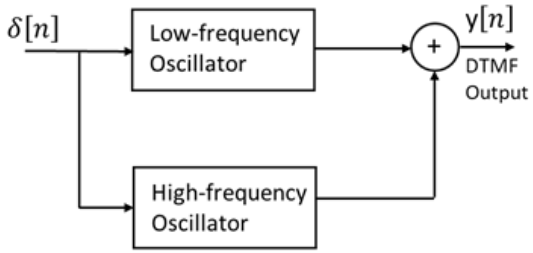


Figure B



Digital Resonator Fundamentals : Digital Oscillators

Example 3:

- A digital sinusoidal oscillator is shown in Figure C :

(a) Assuming θ_0 is the resonant frequency of the digital oscillator, find the values of b_1 and b_2 for sustaining the oscillation.

Answer:

$$H(z) = \frac{K}{1+b_1 z^{-1}+b_2 z^{-2}} \quad \text{----(1)}$$

$$H(z) = \frac{K\ z^2}{(z - re^{j\theta_0})(z - re^{-j\theta_0})} = \frac{Kz^2}{z^2 - 2r\cos\theta_0\, z + r^2} \quad \text{---(2)}$$

By equating (1) and (2), we obtain: $b_1 = -2\, r\cos\theta_0;\ \ b_2 = r^2$

However, for oscillation (marginally stable system) , $b_2 = 1$, $\therefore\ b_1 = -2\cos\theta_0$

(b)

(i) Write the difference equation corresponding to Figure C. You may assume that $x[n] = (A\sin\theta_0)\delta[n]$, and $y(-1) = y(-2) = 0$.

$$y[n] = -b_1 y[n-1] - y[n-2] + x[n]$$

$$y[n] = 2\cos\theta_0\, y[n-1] - y[n-2] + A\sin\theta_0\, \delta[n]$$

(ii) Show, by analysing the difference equation, that the application of an impulse at n = 0 serves the purpose of beginning the sinusoidal oscillation and prove that the oscillation is self-sustaining thereafter.

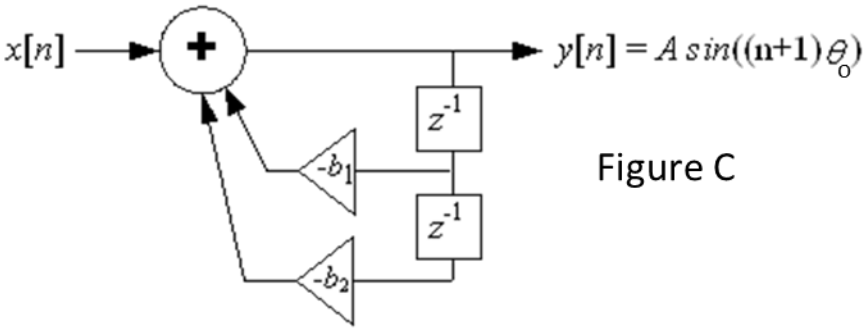


Figure C

Digital Resonator Fundamentals : Digital Oscillators

Example 3:

$$y[n] = 2\cos\theta_0\, y[n-1] - y[n-2] + A\sin\theta_0\, \delta[n]$$

Let n = 0

$y[0] = 2\cos\theta_0\, y[-1] - y[-2] + A\sin\theta_0\, \delta[0]$ {$y[-1] = y[-2] = 0$; $\delta[0]=1$}

$\boldsymbol{y[0] = A\sin\theta_0}$

Let n = 1

$y[1] = 2\cos\theta_0\, y[0] - y[-1] + A\sin\theta_0\, \delta[1]$ {$y[-1] = 0$; $\delta[1]=0$}

$\boldsymbol{y[1] = 2\cos\theta_0\, A\sin\theta_0 = A\sin 2\theta_0}$

Let n = 2

$y[2] = 2\cos\theta_0\, y[1] - y[0] + A\sin\theta_0\, \delta[2]$ {$\delta[2] = 0$}

$= 2\cos\theta_0\, A\sin 2\theta_0 - A\sin\theta_0 = 2A\cos\theta_0\,[2\sin\theta_0\cos\theta_0] - A\sin\theta_0$

$= A\sin\theta_0\,[4\cos^2\theta_0 - 1] = A[3\sin\theta_0 - 4\sin^3\theta_0]$

$\boldsymbol{y[2] = A\sin 3\theta_0}$ and so forth.

$\therefore\ \boldsymbol{y[n] = A\sin(n+1)\theta_0}$ {The oscillation is self sustaining}

(c) By setting the input to zero and under certain initial conditions, sinusoidal oscillation can be obtained using the structure shown in Figure C. Find these initial conditions.

$$y[n] = 2\cos\theta\, y[n-1] - y[n-2] + x[n] \quad \{x[n] = 0 \text{ for an oscillator}\}$$

$$y[n] = 2\cos\theta\, y[n-1] - y[n-2] \ \boldsymbol{and}\ \boldsymbol{y[n] = A\sin(n+1)\theta_0}$$

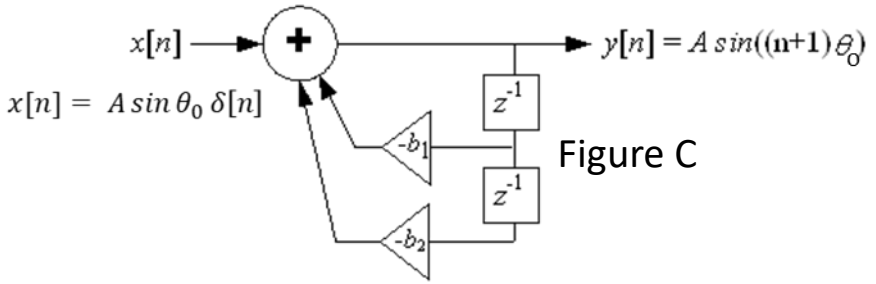


Figure C

Substitute n =-1 in $y[n] = A\sin(n+1)\theta_0$;
$\therefore$ y[-1] =0 - First initial conditions.

Substitute n = -2 in $y[n] = A\sin(n+1)\theta_0$,
$\therefore$ y[-2] =$-A\sin\theta_0$ – Second initial conditions.

Digital Resonator Fundamentals : Digital Oscillators

Sine and Cosine Oscillators

- Sinusoidal oscillators are normally used for generating carrier frequencies in modulation schemes .
- Some modulation schemes require both the sine and cosine oscillators. These signals can be generated simultaneously from the so-called coupled-form oscillator.
- Consider the trigonometric equation for $\cos(n\theta + \theta)$:

$$\cos(n\theta + \theta) = \cos(n\theta)\cos(\theta) - \sin(n\theta)\sin(\theta)$$

Let $\boldsymbol{y[n]} = \boldsymbol{cos}[\boldsymbol{n}\theta]$

$$\therefore y[n+1] = \cos(\theta)\, y[n] - \sin(\theta)\, x[n]$$

Replace *n* by *n*-1:

$$y[n] = \cos(\theta)\, y[n-1] - \sin(\theta)\, x[n-1] \quad \text{------- (1)}$$

- Similarly consider trigonometric equation for $\sin(n\theta + \theta)$

$$\sin(n\theta + \theta) = \sin(n\theta)\cos(\theta) + \cos(n\theta)\sin(\theta)$$

Let $\boldsymbol{x[n]} = \boldsymbol{sin}[\boldsymbol{n}\theta]$

$$\therefore x[n+1] = \sin(\theta)\, y[n] + x[n]\cos(\theta)$$

Replace *n* by *n*-1:

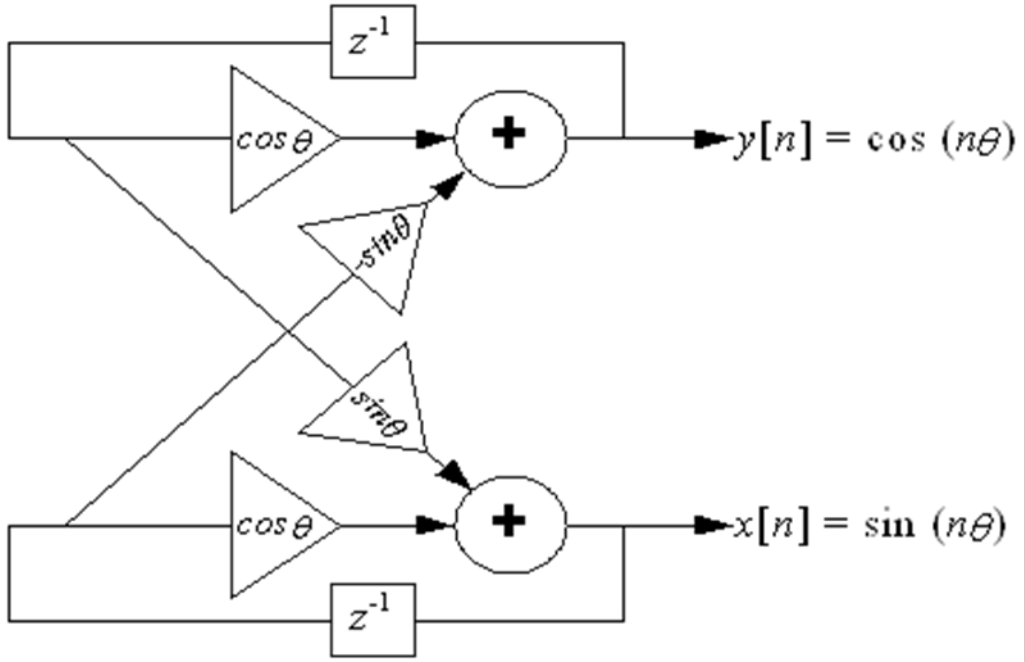
$$x[n] = \sin(\theta)\, y[n-1] + x[n-1]\cos(\theta) \quad \text{-------- (2)}$$

- Thus, we obtain the two coupled difference equations:

$$y[n] = \cos(\theta)\, y[n-1] - \sin(\theta)\, x[n-1] \quad \text{--- (1)}$$

$$x[n] = \sin(\theta)\, y[n-1] + x[n-1]\cos(\theta) \quad \text{--- (2)}$$

- Using equations (1) and (2), the structure for the realisation of the coupled-form oscillator is shown below:



Digital Resonator Fundamentals : Digital Oscillators

Example 4:

- An oscillator is given by the following coupled difference equations expressed in matrix form.

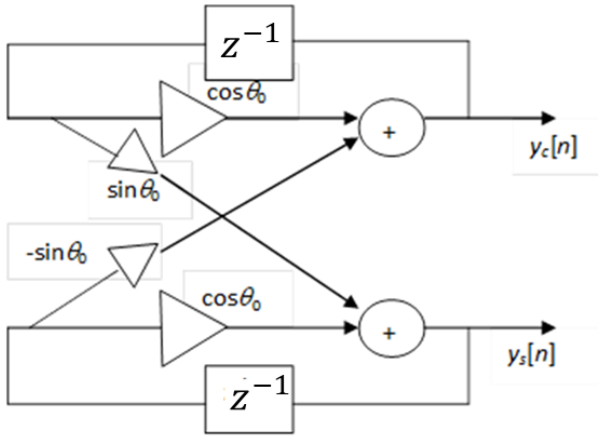
$$\begin{bmatrix} y_c[n] \\ y_s[n] \end{bmatrix} = \begin{bmatrix} cos\,\theta_0 & -sin\,\theta_0 \\ sin\,\theta_0 & cos\,\theta_0 \end{bmatrix} \begin{bmatrix} y_c[n-1] \\ y_s[n-1] \end{bmatrix}$$

(a) Draw the structure for the realisation of this oscillator, where θ_0 is the oscillation frequency.

(b) If the initial conditions $y_c[-1] = A cos\theta_0$ and $y_s[-1] = -A sin\theta_0$, obtain the outputs $y_c[n]$ and $y_s[n]$ using the above difference equations.

Answer:

$$y_c[n] = cos\,\theta_0\, y_c[n-1] - sin\,\theta_0\, y_s[n-1]$$
$$y_s[n] = sin\,\theta_0\, y_c[n-1] + cos\,\theta_0\, y_s[n-1]$$



(b)

$\boldsymbol{Let\ n = 0}$

$y_s[0] = \sin\theta_0\,(A\cos\theta_0) + \cos\theta_0\,(-A sin\theta_0) = 0$

$y_c[0] = \cos\theta_0\,(A cos\theta_0) - \sin\theta_0\,(-A sin\theta_0) = \boldsymbol{A}$

$\boldsymbol{Let\ n = 1}$

$y_c[1] = \cos\theta_0 . A - \sin\theta_0 .0 = \boldsymbol{A cos[1\ \theta_0]}$

$y_s[1] = A\sin\theta_0 + 0 = A\sin\theta_0$

Let $\boldsymbol{n = 2}$

$y_c[2] = \cos\theta_0\, yc[1] - \sin\theta_0\, ys[1]$

$= \cos\theta_0\, A\cos\theta_0 - \sin\theta_0\, A\sin\theta_0$

$= \boldsymbol{A}\cos[\boldsymbol{2}\ \theta_0]$

Let $\boldsymbol{n = n}$

$y_c[n] = \boldsymbol{A}\cos[\boldsymbol{n}\ \theta_0]$

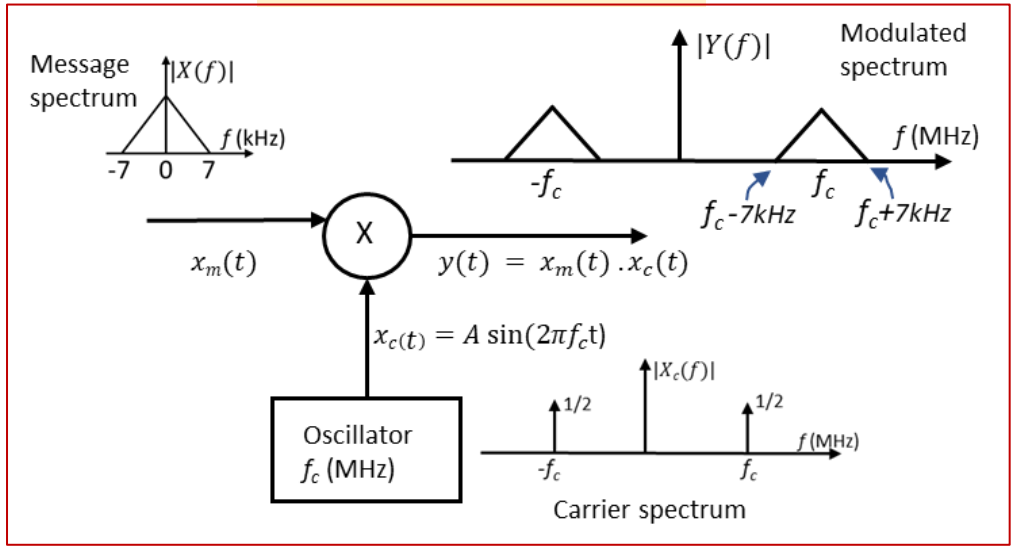
$Similarly \quad y_s[n] = A\sin[n\theta_0]$

Digital Resonator Fundamentals : Digital Oscillators

Note:

- Communication of information often requires transmission of message through space by converting electrical signals to electromagnetic waves using antennas.
- Efficient radiation requires antenna dimensions comparable to the wavelength of the signal .
- The wavelength λ of a radiated signal equals $\frac{c}{f}$, where c is the speed of light (at which electromagnetic waves travel) and f is the signal frequency.
- Transmission of low frequency messages, such as speech and music, would require huge antenna sizes (in km).
- A practical approach is to shift low-frequency messages to a higher frequency using *modulation* and to transmit the modulated signal instead.
- In turn, this requires a means of recovering the original message at the receiving end using a process called *demodulation*.
- Such modulation Scheme (known as amplitude modulation) is shown on the right, where the carrier frequency is generated by an oscillator.

Modulation Scheme



- The amplitude modulation scheme typically modulate the amplitude of a high frequency carrier by a message signal.
- In the above scheme (Double-sideband suppressed carrier), the carrier $x_c(t)$ is multiplied by the message $x_m(t)$.
- The message spectrum is centered about $\pm f_c$ as shown above.
- At the receiver, the message $x_m(t)$ can be recovered using demodulation

ELEC3104: Digital Signal Processing

Topic 4: Digital Resonator Fundamentals

- Second-Order Digital Resonator (pp. 3-8)
- Stability of a Second - Order Filter (pp. 9-11)
- Digital Oscillators (pp. 12-19)
- Tutorial questions (pp. 20-25)

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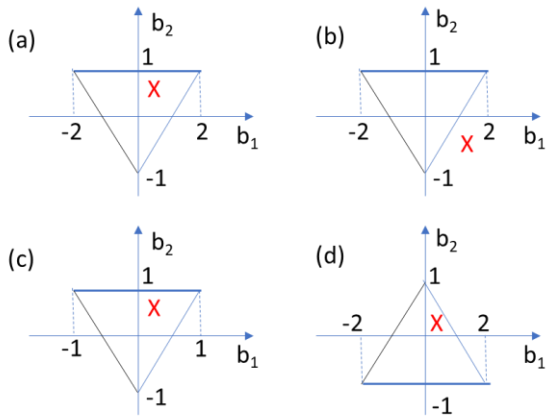
Term 3, 2021

Topic 4: Tutorial Questions (SOLO -Level 1)

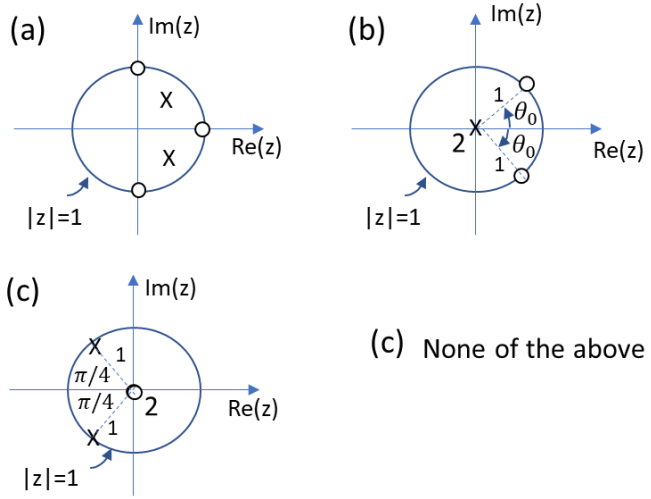
SOLO – Level 1

1. A second-order all-pole filter is described by $H(z) = \frac{1}{1+b_1 z^{-1}+b_2 z^{-2}}$

   Which of the following diagram(s) is used to conclude that the system is stable ? (Tick the correct answer)



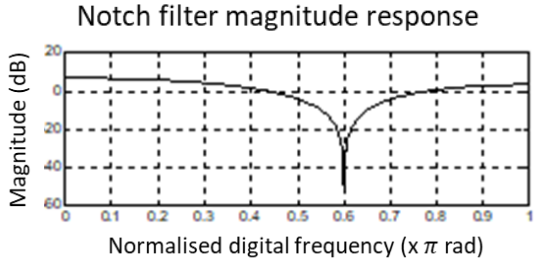
2. Which of the following pole-zero plot represents a digital oscillator? (Tick the correct answer)

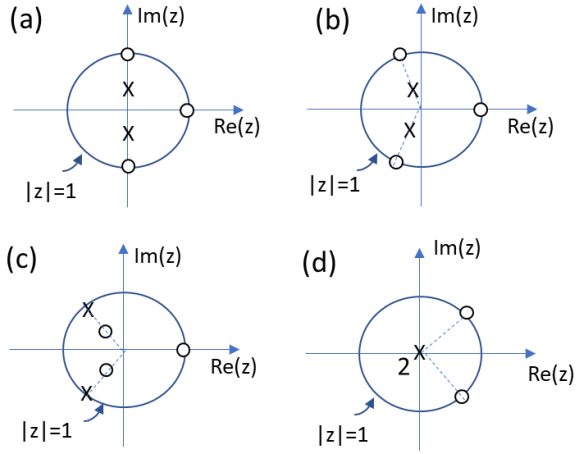


(c) None of the above

SOLO – Level 1

3. Which of the following pole-zero plot is most likely to produce the notch filter response shown below ?(Tick the correct answer)





4. An instrumentation and recording system require that the power-line frequency of 50Hz and harmonics to be eliminated. What type of filter would you choose? (tick the correct answer)

   (a) All-pass filter
   (b) Band-pass filter
   (c) Notch filter
   (d) High-pass filter
   (e) Low- pass filter
   (f) An oscillator
   (g) None of the above

Topic 4: Tutorial Questions (Level 2)

SOLO – Level 2

1. A digital oscillator has a unit impulse response given by:

$$h[n] = \sin([n+1]\theta_0)\, u[n];$$

where θ_0 is the oscillator frequency;

   (a) Find the transfer function of this oscillator
   (b) Draw the structure for this transfer function using the transfer function obtained in part (a) above
   (c) By setting the input in part (b) to zero and under certain initial conditions, sinusoidal oscillation can be obtained. Find these initial conditions.

2. Using the following difference equation, you are required to implement (i.e. draw a structure) a digital oscillator that generates y[n]=sin(θ_0n) for n ≥0. Choose appropriate initial conditions. Note that: $x[n]$ is the input; $y[n]$ is the output and r is a constant. You are required to choose an appropriate value for r in order to sustain oscillation.

$$y[n] = 2r\cos\theta_0\, y[n-1] - r^2 y[n-2] + x[n] - r\cos\theta_0\, x[n-1]$$

3. Design a notch filter with notch frequency 60 Hz, stopband 5 Hz (bandwidth) and sampling frequency 300Hz.

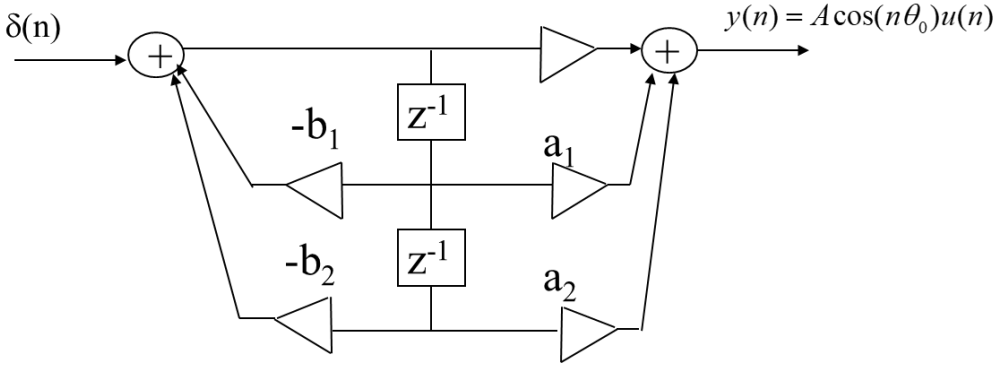
4. Design a bandpass filter with centre frequency of 100Hz, passband 10Hz, stopband edges at 50 Hz and 150 Hz , and sampling frequency 400Hz.
   Note: If the designed filter becomes non causal, then use double poles at each pole locations to get a causal filter.

5. An IIR oscillator which is initialised by an impulse is shown below. The desired impulse response is :

$$y[n] = A\cos[n\theta_0]u[n];$$

where θ_0 is the oscillator frequency;

Assuming θ_0 is the oscillator frequency of the digital oscillator, and by writing appropriate equations, find the values of b_1,b_2, a_0, a_1 and a_2 for sustaining the oscillation.



6. The transfer function of a digital resonator with zeros at the origin is given by

$$H(z) = \frac{b_0}{(1 - re^{j\theta_0}z^{-1})(1 - re^{-j\theta_0}z^{-1})}$$

where r is the pole radius and θ_0 is the resonant frequency.

Show that the gain b_0 is given by the following, when $|H(\theta_0)| = 1$,

$$b_0 = (1-r)\sqrt{1 + r^2 - 2r\cos(2\theta_0)}$$

Topic 4: Tutorial questions (Level 3)

SOLO – Level 3

1. For the structure shown in Figure 1, write down the appropriate difference equations and hence state the function of this structure.

2. The transfer function of a digital resonator with zeros at the origin is given by

$$H(z) = \frac{b_0(1 - z^{-1})(1 + z^{-1})}{(1 - re^{j\theta_0}z^{-1})(1 - re^{-j\theta_0}z^{-1})}$$

where r is the pole radius and θ_0 is the resonant frequency. Determine the gain b_0 for the digital resonator so that $|H(\theta_0)| = 1$.

3. Using the z-transform table, (a) determine the difference equations for the digital oscillators that have impulse responses $h[n] = A\,sin[n\theta_0]u[n]$ and $h[n] = A\,cos[n\theta_0]u[n]$ respectively; (b) Determine the structure for the coupled-form oscillator by combining the structure for the digital oscillator obtained in part (a) above.

4. Design an FIR filter that will reject a very strong 60-Hz sinusoidal interference contaminating a 200Hz useful sinusoidal signal. Determine the gain of the filter so that the useful signal does not change amplitude. The filter operates at a sampling frequency f_s = 500 samples/sec.

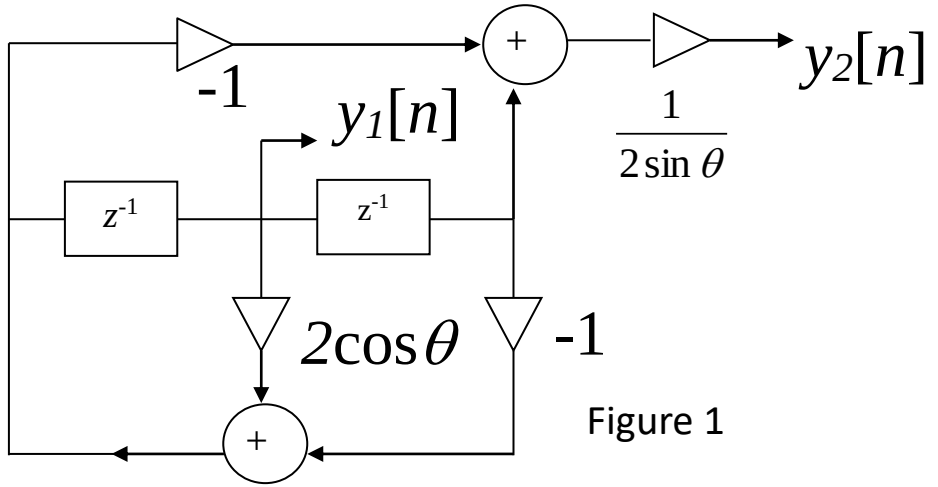


Figure 1

5. The transfer function of a digital resonator is given by

$$H(z) = \frac{b_0}{(1 - re^{j\theta_0}z^{-1})(1 - re^{-j\theta_0}z^{-1})}$$

The impulse response of this resonator can be found by partial fraction expansion . Show that the impulse response h[n] is given by:

$$h[n] = \frac{b_0\, r^n}{\sin(\theta_0)} \sin[(n+1)\theta_0]\, u[n]$$

Topic 4: Tutorial questions (Level 4)

SOLO – Level 4

1. For the system shown in figure 2,

   (a) Express $y_1[n]$ in terms of $y_1[n-1]$, $y_2[n-1]$, and $x[n]$; do the same for $y_2[n]$.

   (b) You may assume $A = \cos(\theta_0)$; $B = \sin(\theta_0)$; $y_1[-1] = \cos(-\theta_0)$; $y_2(-1) = \sin(-\theta_0)$. If $x[n] = 0$, show that : $y_1[n] = \cos(n\theta_0)$ and $y_2[n] = \sin(n\theta_0)$

   (c) Obtain the system transfer function

   $H_1(z) = \frac{Y_1(z)}{X(z)}$ and $H_2(z) = \frac{Y_2(z)}{X(z)}$ where A and B arbitrary values.

   (d) If $A = B = \frac{\sqrt{2}}{2}$, sketch the pole-zero pattern for $H_1(z)$ and $H_2(z)$

   (e) If $A = B = \frac{\sqrt{2}}{2}$ and $x[n] = \delta[n]$, find the impulse response $h_1[n]$.

2. Consider a filter with transfer function

$$H(z) = \frac{b_0\left(1 - e^{j\theta_0}z^{-1}\right)\left(1 - e^{-j\theta_0}z^{-1}\right)}{\left(1 - re^{j\theta_0}z^{-1}\right)\left(1 - re^{-j\theta_0}z^{-1}\right)}$$

   (a) Sketch the pole-zero pattern.

   (b) Show that for $r \approx 1$, , the system is a notch filter and provide a rough sketch of its magnitude response if $\theta_0 = 60^o$.

   (c) For $\theta_0 = 60^o$, choose b_0 so that the maximum value $|H(\theta)| = 1$.

   (d) Determine the approximate 3-dB bandwidth of this filter.

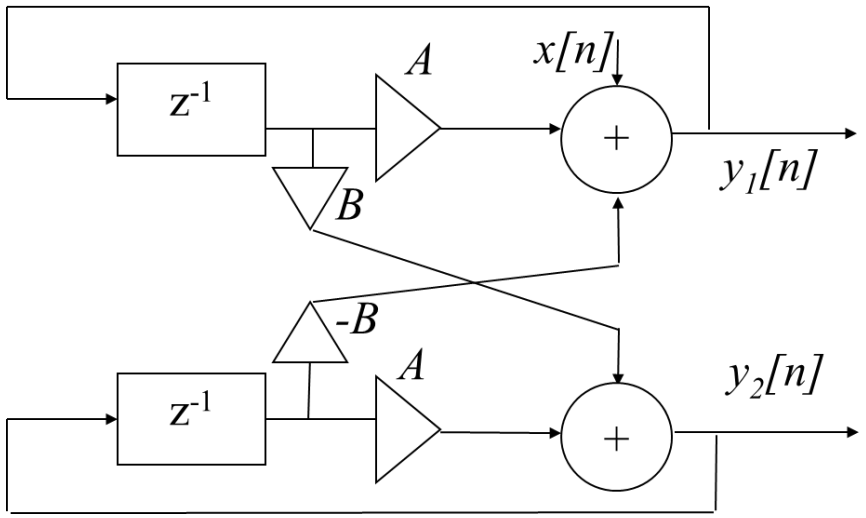


Figure 2

3. By calculating the magnitude response of the following filter with a transfer function

$$H(z) = 1 + z^{-1} + z^{-2} + \ldots\ldots + z^{-8}$$

   determine the frequencies of the analogue sinusoids that cannot pass through the filter. You may assume the sampling frequency $f_s = 1$ kHz.

Topic 4: Tutorial questions (Level 5)

SOLO – Level 5

1. Design a digital oscillator with adjustable phase, $\emptyset$, and produces an output

$$y[n] = \cos(n\theta_0 + \emptyset)\, u[n]$$

2. A digital notch filter is required to remove an undesirable 60 Hz hum associated with a power supply in an ECG recording application. The sampling frequency used is fs = 500 samples/sec.

   (a) Design a second-order FIR notch filter and

   (b) A second-order pole-zero notch filter for this purpose

   In both cases choose the gain b_0 so that $|H(\theta)| = 1\ for\ \theta = 0$.

3. A comb filter may be viewed as a notch filter in which the spectral nulls occur periodically across the frequency band. Comb filters find applications in a wide range of practical systems, for example, in the rejection of power-line harmonics. A comb filter can be described using a moving average FIR filter with the difference equation

$$y[n] = \frac{1}{M+1}\sum_{k=0}^{M} x[n-k]$$

   (a) Show that the frequency response of the above filter is given by $H(\theta) = \frac{e^{-j\theta M/2}}{M+1}\frac{\sin\theta(\frac{M+1}{2})}{\sin(\frac{\theta}{2})}$

   (b) Sketch the magnitude characteristic of the comb filter (for M= 10) that you obtained in part (a) above, but clearly illustrating the existence of the periodically spaced zeros at $\theta_k = \frac{2\pi k}{M+1}\ for\ k = 1,2,\ldots.M$.